AP Seminar

Evidence Guide

Evidence: Experimentation, Examples, Authoritative Testimony, Statistics

Different disciplines use different kinds of evidence:

- In literary studies, the texts are usually the chief evidence.
- In the social sciences, field research (interviews, surveys) usually provides evidence.
- In the sciences, reports of experiments are the usual evidence; if an assertion cannot be tested - if one cannot show it to be false - it is a belief, an opinion, not a scientific hypothesis.

EXPERIMENTATION

Induction is obviously useful in arguing. If, for example, one is arguing that handguns should be controlled, one will point to specific cases in which handguns caused accidents or were used to commit crimes. In arguing that abortion has a traumatic effect on women, one will point to women who testify to that effect. Each instance constitutes evidence for the relevant generalization.

In a courtroom, evidence bearing on the guilt of the accused is introduced by the prosecution, and evidence to the contrary is introduced by the defense. Not all evidence is admissible (e.g., hearsay is not, even if it's true), and the law of evidence is a highly developed subject in jurisprudence. In the forum of daily life, the sources of evidence are less disciplined. Daily experience, a particularly memorable observation, an unusual event - any or all of these may serve as evidence for (or against) some belief, theory, hypothesis, or explanation. Science involves the systematic study of what experience can yield, and one of the most distinctive features of the evidence that scientists can marshal on behalf of their claims is that it is the result of experimentation. Experiments are deliberately contrived situations, often complex in their technology, that are designed to yield particular observations. What the ordinary person does with unaided eye and ear, the scientist does, much more carefully and thoroughly, with the help of laboratory instruments. The variety, extent, and reliability of the evidence obtained in daily life are quite different from those obtained in the laboratory. It's no surprise that society attaches much more weight to the "findings" of scientists than to the corroborative (much less the contrary) experiences of ordinary people. No one today would seriously argue that the sun really does go around the earth just because it looks that way; nor would we argue that because viruses are invisible to the naked eye they cannot cause symptoms such as swellings and fevers, which are plainly evident.

EXAMPLES

One form of evidence is the example. Suppose we argue that a candidate is untrustworthy and shouldn't be elected to public office. We point to episodes in his career- his misuse of funds in 2008 and the false charges he made against an opponent in 2016 - as examples of his untrustworthiness. Or if we're arguing that President Truman ordered the atom bomb dropped to save American (and, for that matter, Japanese) lives that otherwise would have been lost in a hard-fought invasion of Japan, we point to the stubbornness of the Japanese defenders in battles on the islands of Saipan, Iwo Jima, and Okinawa, where Japanese soldiers fought to the death rather than surrender.

These examples, we say, indicate that the Japanese defenders of the main islands would have fought to their deaths without surrendering, even though they knew defeat was certain. Or if we argue that the war was nearly won when Truman dropped the bomb, we can cite secret peace feelers as examples of the Japanese willingness to end the war. An example is a sample. These two words come from the same Old French word, *essample*, from the Latin exemplum, which means "something taken out" - that is, a selection from the group. A Yiddish proverb shrewdly says, "For example' is no proof;' but the evidence of well- chosen examples can go a long way toward helping a writer to convince an audience.

In arguments, three sorts of examples are especially common:

- real events
- invented instances (artificial or hypothetical cases)
- analogies

We will treat each of these briefly.

Real Events

In referring to Truman's decision to drop the atom bomb, we've already touched on examples drawn from real events - the battles at Saipan and elsewhere. And we've also seen Ben Franklin pointing to an allegedly real happening, a fish that had consumed a smaller fish. The advantage of an example drawn from real life, whether a great historical event or a local incident, is that its reality gives it weight. It cannot simply be brushed off.

Yet an example drawn from reality may not be as clear-cut as we would like. Suppose, for instance, that someone cites the Japanese army's behavior on Saipan and on Iwo Jima as evidence that the Japanese later would have fought to the death in an American invasion of Japan and would therefore have inflicted terrible losses on themselves and on the Americans. This example is open to the response that in June and July 1945 certain Japanese diplomats sent out secret peace feelers, so that in August 1945, when Truman authorized dropping the bomb, the situation was very different.

Similarly, in support of the argument that nations will no longer resort to using atomic weapons, some people have offered as evidence the fact that since World War I the great powers have not used poison gas. But the argument needs more support than this fact provides. Poison gas wasn't decisive or even highly effective in World War 1. Moreover, the invention of gas masks made its use obsolete.

In short, any real event is so entangled in historical circumstances that it might not be adequate or relevant evidence in the case being argued. In using a real event as an example (a perfectly valid strategy), the writer must demonstrate that the event can be taken out of its historical context for use in the new context of argument. Thus, in an argument against using atomic weapons in warfare, the many deaths and horrible injuries inflicted on the Japanese at Hiroshima and Nagasaki can be cited as effects of nuclear weapons that would invariably occur and did not depend on any special circumstances of their use in Japan in 1945.

Invented Instances

Artificial or **hypothetical cases** - **invented instances** - have the great advantage of being protected from objections of the sort we have just given. Recall Thoreau's trout in the milk; that was a colorful hypothetical case that illustrated his point well. An invented instance ("Let's assume that a burglar promises not to shoot a householder if the householder swears not to identify him. Is the householder bound by the oath?") is something like a drawing of a flower in a botany textbook or a diagram of the folds of a mountain in a geology textbook. It is admittedly false, but by virtue of its simplifications it sets forth the relevant details very clearly. Thus, in a discussion of rights, the philosopher Charles Frankel says:

> Strictly speaking, when we assert a right for *X*, we assert that *Y* has a duty. Strictly speaking, that *Y* has such a duty presupposes that *Y* has the capacity to perform this duty. It would be nonsense to say, for example, that a non-swimmer has a moral duty to swim to the help of a drowning man.

This invented example is admirably clear, and it is immune to charges that might muddy the issue if Frankel, instead of referring to a wholly abstract person, *Y*, talked about some real person, Jones, who did not rescue a drowning man. For then Frankel would get bogged down over arguing about whether Jones really couldn't swim well enough to help, and so on.

Yet invented examples have drawbacks. First and foremost, they cannot serve as evidence. A purely hypothetical example can illustrate a point or provoke reconsideration of a generalization, but it cannot substitute for actual events as evidence supporting an inductive inference. Sometimes, such examples are so fanciful that they fail to convince the reader. Thus, the philosopher Judith Jarvis Thomson, in the course of an argument entitled "A Defense of Abortion," asks the reader to imagine waking up one day and finding that against her will a celebrated violinist whose body is not adequately functioning has been hooked up into her body for life support. Does she have the right to unplug the violinist? As you read the essays we present in this textbook, you'll have to decide for yourself whether the invented cases proposed by various authors are helpful or whether they are so remote that they hinder thought. Readers will have to decide, too, about when they can use invented cases to advance their own arguments.

But we add one point: Even a highly fanciful invented case can have the valuable effect of forcing us to see where we stand. A person may say that she is, in all circumstances, against vivisection - the practice of performing operations on live animals for the purpose of research. But what would she say if she thought that an experiment on one mouse would save the life of someone she loves? Conversely, if she approves of vivisection, would she also approve of sacrificing the last giant panda to save the life of a senile stranger, a person who in any case probably wouldn't live longer than another year? Artificial cases of this sort can help us to see that we didn't really mean to say such-and-such when we said so-and-so.

Analogies

The third sort of example, **analogy**, is a kind of comparison. An analogy asserts that things that are alike in some ways are alike in yet another way as well. Here's an example:

> Before the Roman Empire declined as a world power, it exhibited a decline in morals and in physical stamina; our society today shows a decline in both morals (consider the high divorce rate and the crime rate) and physical culture (consider obesity in children). America, like Rome, will decline as a world power.

Strictly speaking, an analogy is an extended comparison in which different things are shown to be similar in several ways. Thus, if one wants to argue that a head of state should have extraordinary power during wartime, one can argue that the state at such a time is like a ship in a storm: The crew is needed to lend its help, but the decisions are best left to the captain. (Notice that an analogy compares things that are relatively unlike.

Comparing the plight of one ship to another or of one government to another isn't an analogy; it's an inductive inference from one case of the same sort to another such case.)

Let's consider another analogy. We have already glanced at Judith Thomson's hypothetical case in which the reader wakes up to find herself hooked up to a violinist in need of life support. Thomson uses this situation as an analogy in an argument about abortion. The reader stands for the mother; the violinist, for the unwanted fetus. You may want to think about whether this analogy is close enough to pregnancy to help illuminate your own thinking about abortion.

The problem with argument by analogy is this: Two admittedly different things are agreed to be similar in several ways, and the arguer goes on to assert or imply that they are also similar in another way- the point being argued. (That's why Thomson argues that if something is true of the reader-hooked-up-to-a-violinist, it is also true of the pregnant-mother-hooked- up-to-a-fetus.) But the two things that are said to be analogous and that are indeed similar in characteristics A, B, and C are also different -let's say in characteristics D and E. As Bishop Butler is said to have remarked in the early eighteenth century, "Everything is what it is, and not another thing."

Analogies can be convincing, especially because they can make complex issues seem simple. "Don't change horses in midstream" isn't a statement about riding horses across a river but, rather, about choosing new leaders in critical times. Still, in the end, analogies don't necessarily prove anything. What may be true about riding horses across a stream may not be true about choosing new leaders in troubled times. Riding horses across a stream and choosing new leaders are fundamentally different things, and however much they may be said to resemble each other, they remain different. What is true for one need not be true for the other.

Analogies can be helpful in developing our thoughts and in helping listeners or readers to understand a point we're trying to make. It is sometimes argued, for instance - on the analogy of the doctor-patient, the lawyer-client, or the priest-penitent relationship - that newspaper and television reporters should not be required to reveal their confidential sources. That is worth thinking about: Do the similarities run deep enough, or are there fundamental differences? Consider another example: Some writers who support abortion argue that the fetus is not a person any more than the acorn is an oak. That is also worth thinking about. But one should also think about this response: A fetus is not a person, just as an acorn is not an oak; but an acorn is a potential oak, and a fetus is a potential person, a potential adult human being. Children, even newborn infants, have rights, and one way to explain this claim is to call attention to their potentiality to become mature adults. Thus, some people argue that the fetus, by analogy, has the rights of an infant, for the fetus, like the infant, is a potential adult.

Three analogies for consideration: First, let's examine a brief comparison made by Jill Knight, a member of the British Parliament, speaking about abortion:

> Babies are not like bad teeth, to be jerked out because they cause suffering.

Her point is effectively put; it remains for the reader to decide whether fetuses are babies and if a fetus is not a baby, why it can or cannot be treated like a bad tooth.

Now a second bit of analogical reasoning, again about abortion: Thomas Sowell, an economist at the Hoover Institute, grants that women have a legal right to abortion, but he objects to a requirement that the government pay for abortions:

> Because the courts have ruled that women have a legal right to an abortion, some people have jumped to the conclusion that the government has to pay for it. You have a constitutional right to privacy, but the government has no obligation to pay for your window shades. (Pink and Brown People, 1981, p. 57)

We leave it to you to decide whether the analogy is compelling - that is, if the points of resemblance are sufficiently significant to allow you to conclude that what's true of people wanting window shades should be true of people wanting abortions.

And one more: A common argument on behalf of legalizing gay marriage drew an analogy between gay marriage and interracial marriage, a practice that was banned in sixteen states until 1967, when the Supreme Court declared miscegenation statutes unconstitutional. The gist of the analogy was this: Racism and discrimination against gay and lesbian people are the same. If marriage is a fundamental right - as the Supreme Court held in its 1967 decision striking down bans on miscegenation - then it is a fundamental right for gay and lesbian people as well as heterosexual people.

AUTHORITATIVE TESTIMONY

Another form of evidence is testimony, the citation or quotation of authorities. In daily life, we rely heavily on authorities of all sorts: We get a doctor's opinion about our health, we read a book because an intelligent friend recommends it, we see a movie because a critic gave it a good review, and we pay at least a little attention to the weather forecaster.

In setting forth an argument, one often tries to show that one's view is supported by notable figures - perhaps Jefferson, Lincoln, Martin Luther King [r., or scientists who won the Nobel Prize. You may recall that in Chapter 2, in talking about medical marijuana legalization, we presented an essay by Sanjay Gupta. To make certain that you were impressed by his ideas, we described him as CNN's chief medical correspondent and a leading public health expert. In our Chapter 2 discussion of Sally Mann, we qualified our description of her controversial photographs by noting that Time magazine called her "America's Best Photographer" and the New Republic called her book "one of the great photograph books of our time." But heed some words of caution:

- Be sure that the authority, however notable, is an authority on the topic in question. (A well-known biologist might be an authority on vitamins but not on the justice of war.)

- Be sure that the authority is unbiased. (A chemist employed by the tobacco industry isn't likely to admit that smoking may be harmful, and a producer of violent video games isn't likely to admit that playing those games stimulates violence.)

- Beware of nameless authorities: "a thousand doctors," "leading educators," "researchers at a major medical school," (If possible, offer at least one specific name.)

- Be careful when using authorities who indeed were great authorities in their day but who now may be out of date. (Examples would include Adam Smith on economics, Julius Caesar on the art of war, Louis Pasteur on medicine).

- Cite authorities whose opinions your readers will value. (William F. Buckley Jr.'s conservative/libertarian opinions mean a good deal to readers of the magazine that he founded, the National Review, but probably not to most liberal thinkers. Gloria Steinem's liberal/feminist opinions carry weight with readers of the magazines that she cofounded, New York and Ms. magazine, but probably not with most conservative thinkers.) When writing for the general reader-your usual audience - cite authorities whom the general reader is likely to accept.

One other point: You may be an authority. You probably aren't nationally known, but on some topics you might have the authority of personal experience. You may have been injured on a motorcycle while riding without wearing a helmet, or you may have escaped injury because you wore a helmet. You may have dropped out of school and then returned. You may have tutored a student whose native language isn't English, you may be such a student who has received tutoring, or you may have attended a school with a bilingual education program. In short, your personal testimony on topics relating to these issues may be invaluable, and a reader will probably consider it seriously.

STATISTICS

The last sort of evidence we discuss here is quantitative, or statistical. The maxim "More is better" captures a basic idea of quantitative evidence: Because we know that 90 percent is greater than 75 percent, we're usually ready to grant that any claim supported by experience in 90 percent of cases is more likely to be true than an alternative claim supported by experience in only 75 percent of cases. The greater the difference, the greater our confidence. Consider an example. Honors at graduation from college are often computed on the basis of a student's cumulative grade-point average (GPA). The undisputed assumption is that the nearer a student's GPA is to a perfect record (4.0), the better scholar he or she is and therefore the more deserving of highest honors. Consequently, a student with a GPA of 3.9 at the end of her senior year is a stronger candidate for graduating summa cum laude than another student with a GPA of 3.6. When faculty members on the honors committee argue over the relative academic merits of graduating seniors, we know that these quantitative, statistical differences in student GPAs will be the basic (if not the only) kind of evidence under discussion.

Graphs, Tables, Numbers

Statistical information can be presented in many forms, but it tends to fall into two main types: the graphic and the numerical. Graphs, tables, and pie charts are familiar ways of presenting quantitative data in an eye-catching manner. To prepare the graphics, however, one first has to decide how best to organize and interpret the numbers, and for some purposes it may be more appropriate to directly present the numbers themselves.

But is it better to present the numbers in percentages or in fractions? Should a report say that the federal budget (1) underwent a twofold increase over the decade; (2) increased by 100 percent; (3) doubled; or (4) at the beginning of the decade was one – half what it was at the end? These are equivalent ways of saying the same thing. Making a choice among them, therefore, will likely rest on whether one's aim is to dramatize the increase (a 100 percent increase looks larger than a doubling) or to play down its size.

Thinking about Statistical Evidence

Statistics often get a bad name because it's so easy to misuse them (unintentionally or not) and so difficult to be sure that they were gathered correctly in the first place. (One old saying goes, "There are lies, damned lies, and statistics") Every branch of social science and natural science needs statistical information, and countless decisions in public and private life are based on quantitative data in statistical form. It's important, therefore, to be sensitive to the sources and reliability of the statistics and to develop a healthy skepticism when you confront statistics whose parentage is not fully explained.

Consider statistics that pop up in conversations about wealth distribution in the United States. In 2014, the Census Bureau calculated that the median household income in the United States was $53,657, meaning that half of households earned less than this amount and half earned above it. However, the average - technically, the mean - household income in the same year was $72,641, about $19,000 (or 39 percent) higher. Which number more accurately represents the typical household income? Both are "correct:' but both are calculated with different measures, median and mean. If a politician wanted to argue that the United States has a strong middle class, he might use the average (mean) income as evidence, a number calculated by dividing the total income of all households by the total number of households. If another politician wished to make a rebuttal, she could point out that the average income paints a rosy picture because the wealthiest households skew the average higher. The median income (representing the number above and below which two halves of all households fall) should be the measure we use, the rebutting politician could argue, because it helps reduce the effect of the limitless ceiling of higher incomes and the finite floor of lower incomes at zero.

Consider the following statistics: Suppose in a given city in 2014, 1 percent of the victims in fatal automobile accidents were bicyclists. In the same city in 2015, the percentage of bicyclists killed in automobile accidents was 2 percent. Was the increase 1 percent (not an alarming figure), or was it 100 percent (a staggering figure)? The answer is both, depending on whether we're comparing (1) bicycle deaths in automobile accidents with all deaths in automobile accidents (that's an increase of 1 percent), or (2) bicycle deaths in automobile accidents only with other bicycle deaths in automobile accidents (an increase of 100 percent). An honest statement would say that bicycle deaths due to automobile accidents doubled in 2015, increasing from 1 to 2 percent. But here's another point: Although every such death is lamentable, if there was one such death in 2014 and two in 2015, the increase from one death to two (an increase of 100 percent!) hardly suggests a growing problem that needs attention. No one would be surprised to learn that in the next year there were no deaths at all, or only one or two.

If it's sometimes difficult to interpret statistics, it's often at least equally difficult to establish accurate statistics. Consider this example:

> Advertisements are the most prevalent and toxic of the mental pollutants. From the moment your radio alarm sounds in the morning to the wee hours of late-night TV, microjolts of commercial pollution flood into your brain at the rate of about three thousand marketing messages per day. (Kalle Lasn, Culture Jam [1999], 18-19)

Lasn's book includes endnotes as documentation, so, being curious about the statistics, we turn to the appropriate page and find this information concerning the source of his data:

> "three thousand marketing messages per day:' Mark Landler, Walecia Konrad, Zachary Schiller, and Lois Therrien, "What Happened to Advertising?" BusinessWeek, September 23, 1991, page 66. Leslie Savan in The Sponsored Life (Temple University Press, 1994), page 1, estimated that "16,000 ads flicker across an individual's consciousness daily:' I did an informal survey in March 1995 and found the number to be closer to 1,500 (this included all marketing messages, corporate images, logos, ads, brand names, on TV, radio, billboards, buildings, signs, clothing, appliances, in cyberspace, etc., over a typical twenty-four hour period in my life). (219)

Well, this endnote is odd. In the earlier passage, the author asserted that about "three thousand marketing messages per day" flood into a person's brain. In the documentation, he cites a source for that statistic from Business Week - though we haven't the faintest idea how the authors of the Business Week article came up with that figure. Oddly, he goes on to offer a very different figure (16,000 ads) and then, to our confusion, offers yet a third figure, 1,500, based on his own "informal survey:'

Probably the one thing we can safely say about all three figures is that none of them means very much. Even if the compilers of the statistics explained exactly how they counted -let's say that among countless other criteria they assumed that the average person reads one magazine per day and that the average magazine contains 124 advertisements - it would be hard to take them seriously. After all, in leafing through a magazine, some people may read many ads and some may read none. Some people may read some ads carefully - but perhaps just to enjoy their absurdity. Our point: Although the author in his text said, without implying any uncertainty, that "about three thousand marketing messages per day" reach an individual, it's evident (by checking the endnote) that even he is confused about the figure he gives.

Unreliable Statistics

We'd like to make a final point about the unreliability of some statistical information - data that looks impressive but that is, in fact, insubstantial. For instance, Marilyn Jager Adams studied the number of hours that families read to their children in the five or so years before the children start attending school. In her book *Beginning to Read: Thinking and Learning about Print* (1994), she pointed out that in all those preschool years, poor families read to their children only 25 hours, whereas in the same period middle- income families read 1,000 to 1,700 hours. The figures were much quoted in newspapers and by children's advocacy groups. Adams could not, of course, interview every family in these two groups; she had to rely on samples. What were her samples? For poor families, she selected 24 children in 20 families, all in Southern California. Ask yourself: Can families from only one geographic area provide an adequate sample for a topic such as this? Moreover, let's think about Adams's sample of middle-class families. How many families constituted that sample? Exactly one - her own. We leave it to you to judge the validity of her findings.

A CHECKLIST FOR EVALUATING STATISTICAL EVIDENCE

Regard statistical evidence (like all other evidence) cautiously, and don't accept it until you have thought about these questions:

- Was it compiled by a disinterested (impartial) source? The source's name doesn't always reveal its particular angle (e.g., People for the American Way), but sometimes it lets you know what to expect (e.g., National Rifle Association, American Civil Liberties Union).
- Is it based on an adequate sample?
- Is the statistical evidence recent enough to be relevant?
- How many of the factors likely to be relevant were identified and measured?
- Are the figures open to a different and equally plausible interpretation?
- If a percentage is cited, is it the average (or mean), or is it the median?

We are not suggesting that everyone who uses statistics is trying to deceive or is unconsciously being deceived by them. We suggest only that statistics are open to widely different interpretations and that often those columns of numbers, which appear to be so precise with their decimal points, may actually be imprecise and possibly worthless if they're based on insufficient or biased samples.